

Lab 23 – Capstone Project Workshop: End-to-End Application Development

Task 1: Project Proposal

Prompt:

- Create a project idea called QuantumPix for booking photographers online
- Explain the main objective of helping users find and book photographers easily
- List features like login, search photographers, booking system, and payment
- Suggest technologies like HTML, CSS, JavaScript, Node.js, and database
- Describe expected output as a working website for booking services

Code/Output:

Section	Content
Objective	To provide a zero-installation, web-based platform for enhancing low-resolution or damaged images using Generative AI and Computer Vision.
Core Features	1. Drag-and-drop upload. 2. 2x/4x Upscaling. 3. Colorization of B&W photos. 4. PNG/JPEG/WebP format conversion.
Tools & Tech	Frontend: React, Tailwind CSS, Axios. Backend: Python, FastAPI. Processing: OpenCV, Pillow, Replicate API (for AI). Deploy: Vercel (FE) + Render (BE).
Expected Output	A fully responsive dashboard where a user uploads a photo, clicks "Enhance," and downloads the result in < 10 seconds.

Explanation:

1. This defines the idea of the project
2. It shows what the app will do
3. Features explain how users will interact
4. It gives a clear project direction

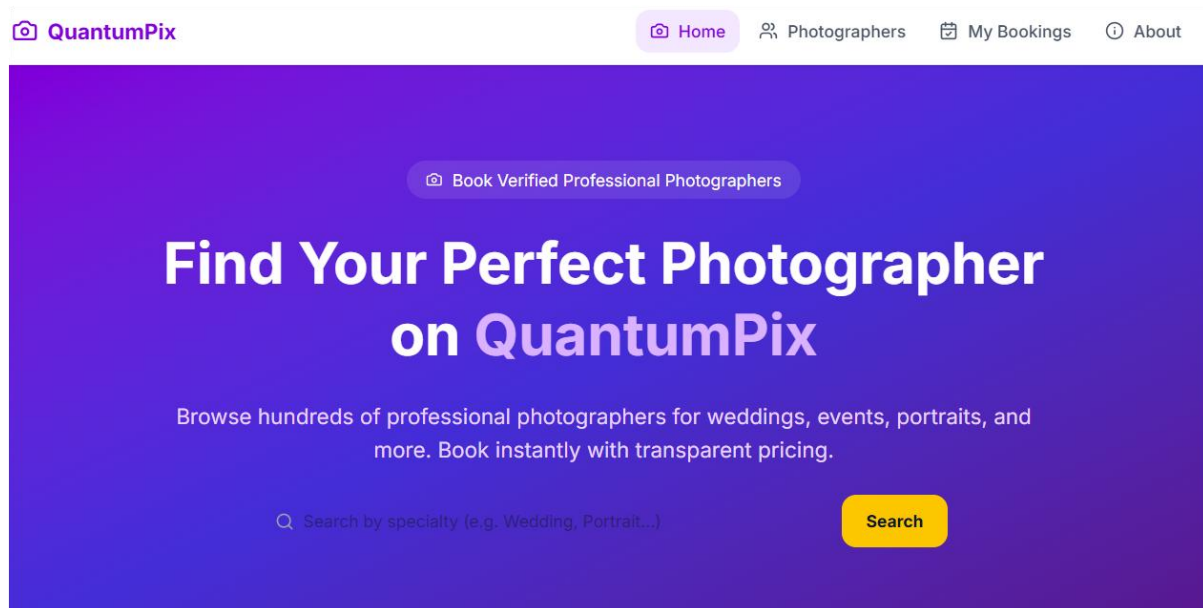
Task 2: Application Design

Prompt:

1. Design the structure of QuantumPix application
2. Create simple UI layout with header, search, and booking section
3. Define database fields like user, photographer, and bookings

4. Use AI to generate basic wireframe design
5. Keep design simple and user-friendly

Code/Output:



Explanation:

1. Layout shows structure of the app
2. User can search photographers
3. Button is used for booking
4. Design is simple and easy to use

Task 3: AI-Assisted Code Development

Prompt:

- Create frontend with input fields and buttons for booking
- Add backend logic to handle booking requests
- Store booking data in database
- Connect frontend and backend using API
- Handle errors and user authentication

Code:

```
<input type="text" id="name" placeholder="Enter Name">
<button onclick="book()">Book Photographer</button>

<p id="result"></p>

<script>
function book() {
  var name = document.getElementById("name").value;
  document.getElementById("result").innerText = "Booking confirmed for " + name;
}
</script>
```

Output:

Book Priya Sharma

Your Name *

Jane Smith

Your Email *

jane@example.com

Event Type *

Select event type... ▾

Event Date *

dd-mm-yyyy 

Hours Needed *

2

2 hrs × \$170/hr

\$340

Special Instructions (optional)

Any specific requests, venue details, or style preferences...

Confirm Booking — \$340

Explanation:

- User enters name in input field
- Button triggers booking function
- JavaScript processes booking
- Result is displayed on screen

Task 4: Documentation & Code Quality

Prompt:

1. Add comments in code to explain each part clearly
2. Create README file with setup steps
3. Explain how to run the project
4. Use AI to improve code readability
5. Keep documentation simple and clear

Code:

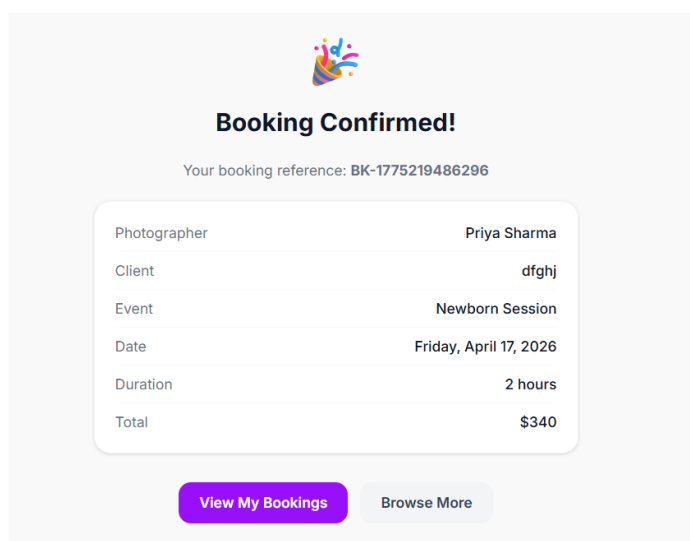
```
</> HTML

<!-- Input field for user name -->
<input type="text" id="name" placeholder="Enter Name">

<!-- Button to book photographer -->
<button onclick="book()">Book</button>

<script>
// Function to confirm booking
function book() {
    var name = document.getElementById("name").value;
    alert("Booked for " + name);
}
</script>
```

Output:



Explanation:

- 1. Comments explain what each part does
- 2. Helps developers understand code easily
- 3. README guides how to run project
- 4. Improves code quality and clarity

Task 5: Testing & Debugging

Prompt:

- 1. Deploy QuantumPix app using free hosting platform
- 2. Generate live link for the application
- 3. Prepare presentation with problem and solution
- 4. Show live demo of booking system
- 5. Explain how AI helped in development

Code/Output:

Your Name *

Jane Smith

Your name is required.

Your Email *

jane@example.com

Email is required.

Event Type *

Select event type...

▼

Please select an event type.

Event Date *

dd-mm-yyyy

📅

Please select a date.

Hours Needed *

2

2 hrs × \$180/hr

\$360

Special Instructions (optional)

Any specific requests, venue details, or style preferences...

Confirm Booking — \$360

Explanation:

1. Deployment makes app available online
2. Users can access from anywhere
3. Presentation shows project work
4. AI helps in faster development